

Lab Worksheet - Exploratory Data Analysis in airquality

Dataset

```
# load libraries:
library(tidyverse) # Recall that tidyverse contains dplyr, ggplot2, and many
other useful packages.

# Load `airquality` dataset:
data("airquality")
airquality |>
  head()
```

	Ozone	Solar.R	Wind	Temp	Month	Day
1	41	190	7.4	67	5	1
2	36	118	8.0	72	5	2
3	12	149	12.6	74	5	3
4	18	313	11.5	62	5	4
5	NA	NA	14.3	56	5	5
6	28	NA	14.9	66	5	6

Q1: Render this quarto file to a pdf. To do so, you can use the hotkey **Ctrl + Shift + K** or **Cmd + Shift + K**. Then, you will see a command appear in the Positron Terminal. Once the command finishes running, you will see a pdf file appear in the same directory as this file and a VIEWER window will display the pdf on the right pane of Positron. Here you should see the rendered contents of this file.

Q2: Convert the **Month** column to a factor with it's levels labeled as month names (e.g. "May", "June", etc.).

```
# your code here
airquality <- airquality |>
  mutate(Month = factor(Month, levels = 5:9, labels = c("May", "June",
"July", "August", "September")))

airquality |>
  head()
```

	Ozone	Solar.R	Wind	Temp	Month	Day
1	41	190	7.4	67	May	1
2	36	118	8.0	72	May	2
3	12	149	12.6	74	May	3
4	18	313	11.5	62	May	4
5	NA	NA	14.3	56	May	5
6	28	NA	14.9	66	May	6

Q3: Create a new column `Temp_C` that converts the temperature from Fahrenheit to Celsius.

```
# your code here
airquality <- airquality |>
  mutate(Temp_C = (Temp - 32) * 5/9)

airquality |>
  head()
```

	Ozone	Solar.R	Wind	Temp	Month	Day	Temp_C
1	41	190	7.4	67	May	1	19.44444
2	36	118	8.0	72	May	2	22.22222
3	12	149	12.6	74	May	3	23.33333
4	18	313	11.5	62	May	4	16.66667
5	NA	NA	14.3	56	May	5	13.33333
6	28	NA	14.9	66	May	6	18.88889

Q4: Rename the `Wind` column to `Wind_mph` and the `Temp` column to `Temp_F`.

```
# your code here
airquality <- airquality |>
  rename(Wind_mph = Wind, Temp_F = Temp)

# or:
# airquality <- airquality |>
#   mutate(Temp_F = Temp, Wind_mph = Wind) |>
#   select(-Temp, -Wind)

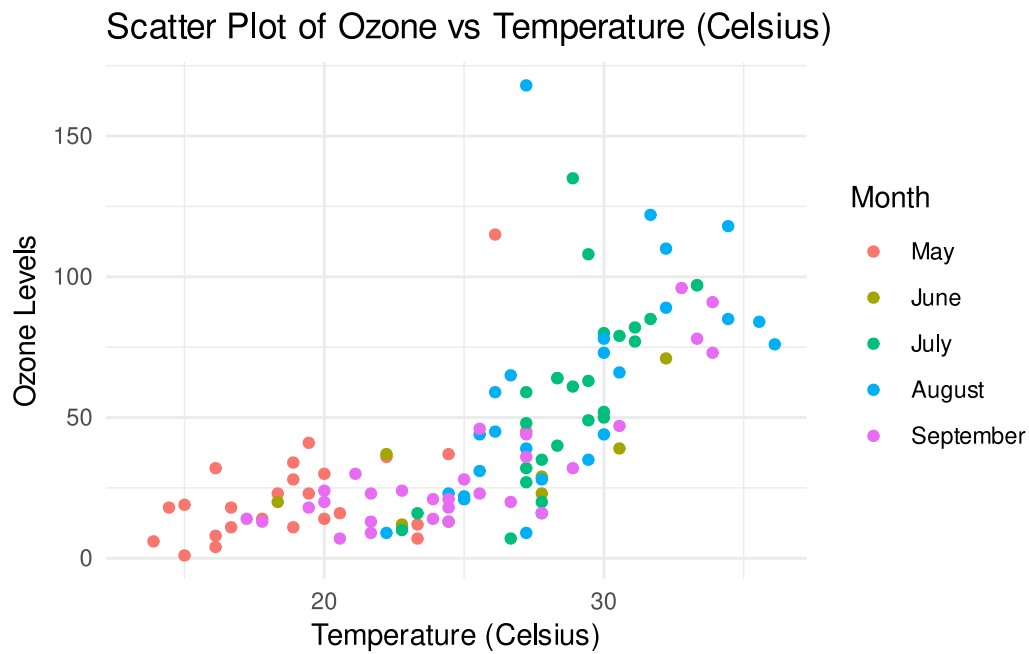
airquality |>
  head()
```

	Ozone	Solar.R	Wind_mph	Temp_F	Month	Day	Temp_C
1	41	190	7.4	67	May	1	19.44444
2	36	118	8.0	72	May	2	22.22222
3	12	149	12.6	74	May	3	23.33333
4	18	313	11.5	62	May	4	16.66667
5	NA	NA	14.3	56	May	5	13.33333
6	28	NA	14.9	66	May	6	18.88889

Q5: Create a scatter plot of `Ozone` vs `Temp_C`, colored by `Month`. Add appropriate axis labels and a title.

```
# your code here
airquality |>
  ggplot(aes(x = Temp_C, y = Ozone, color = Month)) +
  geom_point() +
  labs(title = "Scatter Plot of Ozone vs Temperature (Celsius)",
        x = "Temperature (Celsius)",
        y = "Ozone Levels") +
  theme_minimal()
```

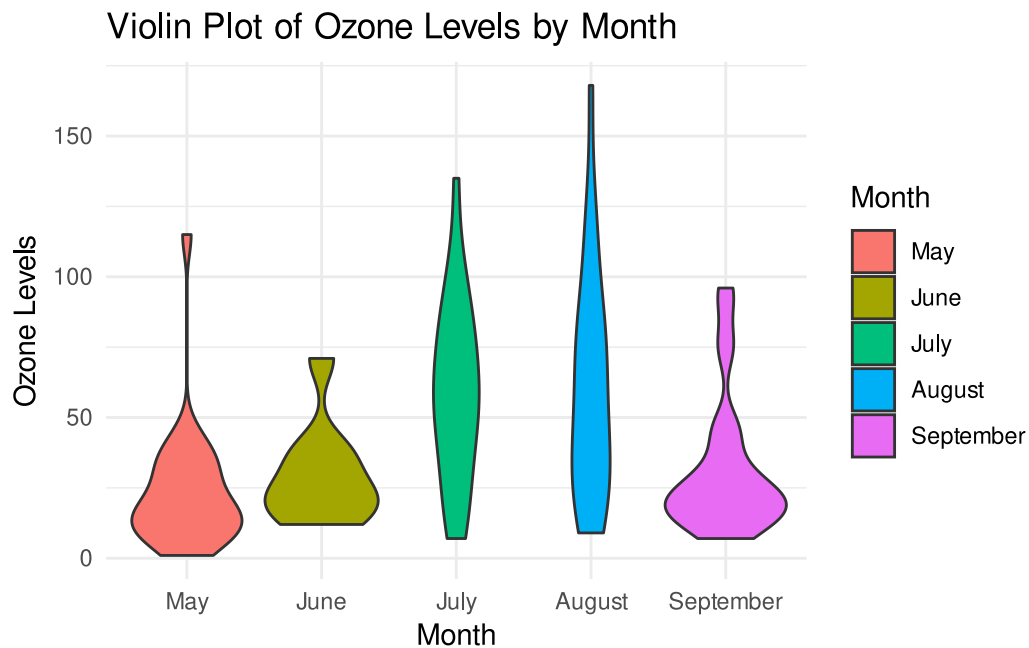
Warning: Removed 37 rows containing missing values or values outside the scale range (``geom_point()``).



Q6: Create a violin plot of `Ozone` levels for each month. Add appropriate axis labels and a title.

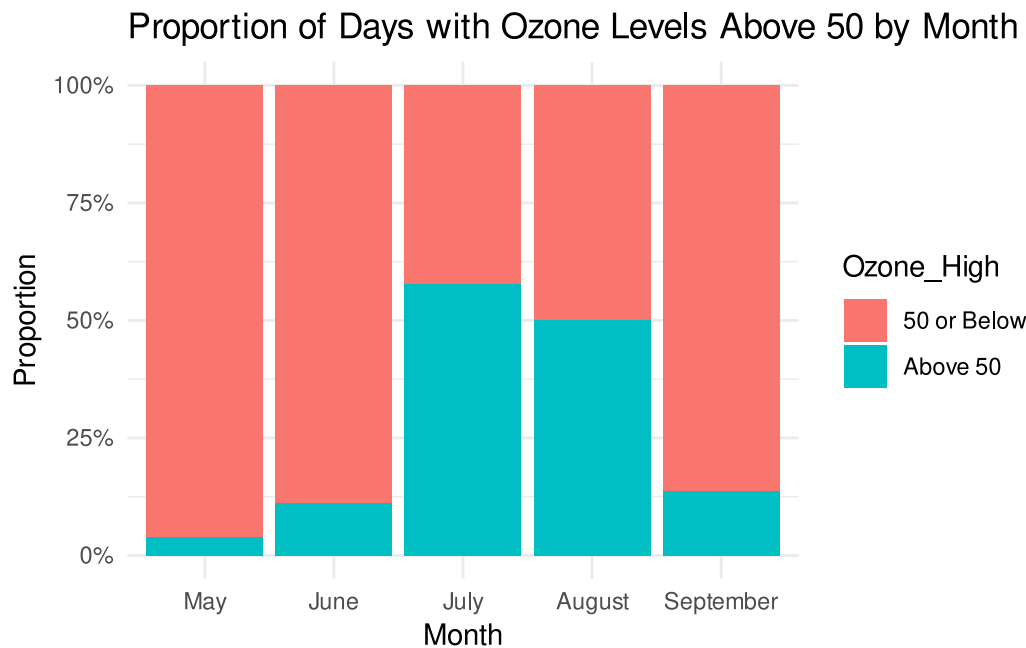
```
# your code here
airquality |>
  ggplot(aes(x = Month, y = Ozone, fill = Month)) +
  geom_violin() +
  labs(title = "Violin Plot of Ozone Levels by Month",
       x = "Month",
       y = "Ozone Levels") +
  theme_minimal()
```

Warning: Removed 37 rows containing non-finite outside the scale range (`'stat_ydensity()'`).

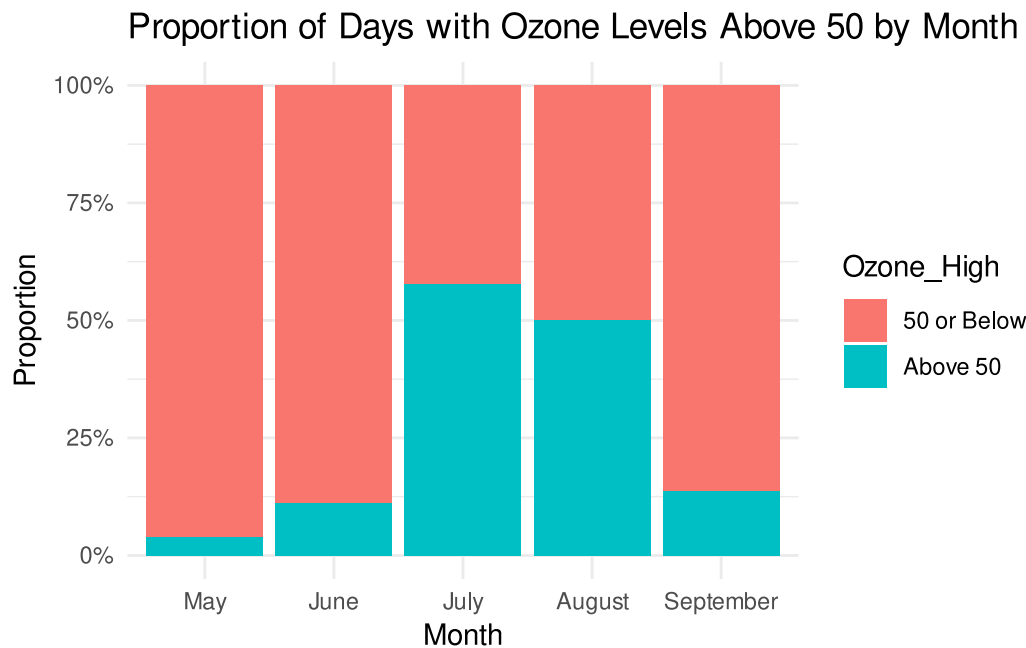


Q7: Create a segmented bar chart of the proportion of days with `Ozone` levels above 50 by `Month`. Remove NA values in the `Ozone` column. Add appropriate axis labels and a title.

```
# your code here
airquality |>
  drop_na(Ozone) |> # or filter(!is.na(Ozone)) |>
  mutate(Ozone_High = ifelse(Ozone > 50, "Above 50", "50 or Below")) |>
  group_by(Month, Ozone_High) |>
  summarise(Count = n(), .groups = 'drop') |>
  group_by(Month) |>
  mutate(Proportion = Count / sum(Count)) |>
  ggplot(aes(x = Month, y = Proportion, fill = Ozone_High)) +
  geom_bar(stat = "identity", position = "fill") +
  labs(title = "Proportion of Days with Ozone Levels Above 50 by Month",
       x = "Month",
       y = "Proportion") +
  scale_y_continuous(labels = scales::percent) +
  theme_minimal()
```

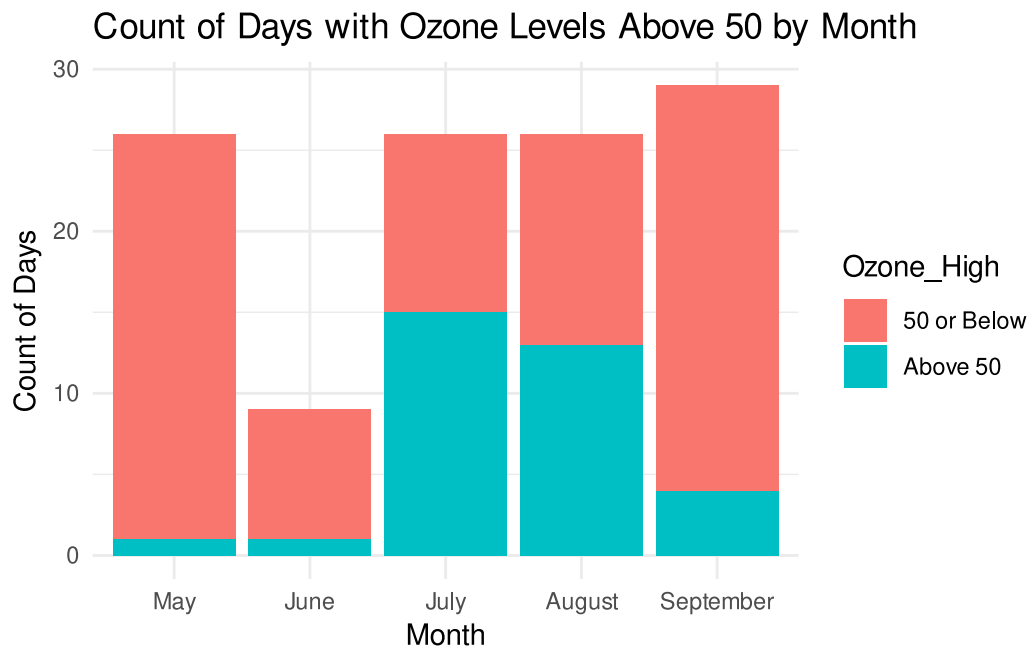


```
# or simply:
airquality |>
  drop_na(Ozone) |> # or filter(!is.na(Ozone)) |>
  mutate(Ozone_High = ifelse(Ozone > 50, "Above 50", "50 or Below")) |>
  ggplot(aes(x = Month, fill = Ozone_High)) + # count is the default
  # statistic in geom_bar, however we fix with position = "fill"
  geom_bar(position = "fill") + # position = "fill" gives proportions
  # instead
  labs(title = "Proportion of Days with Ozone Levels Above 50 by Month",
        x = "Month",
        y = "Proportion") +
  scale_y_continuous(labels = scales::percent) +
  theme_minimal()
```



Q8: Create a stacked bar chart showing the count of days available for each month. Fill bar colors by `Ozone` levels above or below 50 for each month. Remove NA values in the `Ozone` column. Add appropriate axis labels and a title.

```
# your code here
airquality |>
  drop_na(Ozone) |> # or filter(!is.na(Ozone)) |>
  mutate(Ozone_High = ifelse(Ozone > 50, "Above 50", "50 or Below")) |>
  ggplot(aes(x = Month, fill = Ozone_High)) +
  geom_bar() +
  labs(title = "Count of Days with Ozone Levels Above 50 by Month",
       x = "Month",
       y = "Count of Days") +
  theme_minimal()
```



Q9: What is the utility of using a segmented bar chart vs a regular bar chart in this context? Which do you think is more informative and why?

...

Q10: Create a summary table of the number and proportion of NA values in each column.

```
# your code here
airquality |>
  summarize(
    missing_Ozone = sum(is.na(Ozone)),
    prop_missing_Ozone = mean(is.na(Ozone)),
    missing_SolarR = sum(is.na(Solar.R)),
    prop_missing_SolarR = mean(is.na(Solar.R)),
    missing_Wind = sum(is.na(Wind_mph)),
    prop_missing_Wind = mean(is.na(Wind_mph)),
    missing_Temp = sum(is.na(Temp_F)),
    prop_missing_Temp = mean(is.na(Temp_F)),
    missing_Month = sum(is.na(Month)),
    prop_missing_Month = mean(is.na(Month)),
    missing_Day = sum(is.na(Day)),
    prop_missing_Day = mean(is.na(Day))
  )
```



```
) |>
pivot_longer(everything(), names_to = "Metric", values_to = "Value")
```

```
# A tibble: 12 × 2
  Metric      Value
  <chr>      <dbl>
1 missing_Ozone 37
2 prop_missing_Ozone 0.242
3 missing_SolarR 7
4 prop_missing_SolarR 0.0458
5 missing_Wind 0
6 prop_missing_Wind 0
7 missing_Temp 0
8 prop_missing_Temp 0
9 missing_Month 0
10 prop_missing_Month 0
11 missing_Day 0
12 prop_missing_Day 0
```

```
airquality |>
  summarize(across(everything(), ~ sum(is.na(.)))) |>
  pivot_longer(everything(), names_to = "Column", values_to = "Num_NA")
```

```
# A tibble: 7 × 2
  Column  Num_NA
  <chr>   <int>
1 Ozone    37
2 Solar.R    7
3 Wind_mph    0
4 Temp_F    0
5 Month    0
6 Day    0
7 Temp_C    0
```

```
airquality |>
  summarize(across(everything(), ~ mean(is.na(.)))) |>
  pivot_longer(everything(), names_to = "Column", values_to = "Prop_NA")
```

```
# A tibble: 7 × 2
  Column  Prop_NA
  <chr>   <dbl>
```

1	Ozone	0.242
2	Solar.R	0.0458
3	Wind_mph	0
4	Temp_F	0
5	Month	0
6	Day	0
7	Temp_C	0